

Summary and Key Equations of Denoising Diffusion Implicit Models (DDIM)

1 Introduction

Denoising Diffusion Probabilistic Models (DDPMs) have achieved high-quality image generation without adversarial training, but they require a long sequence of steps to generate a sample. Denoising Diffusion Implicit Models (DDIMs) introduce a non-Markovian forward process, allowing deterministic sampling and significantly accelerating the generation process while maintaining sample quality. The DDIM framework retains the same training objective as DDPMs but provides more efficient inference. A key innovation in DDIMs is the introduction of a hyperparameter τ , which defines a subset of time steps used in the reverse process to speed up sampling.

2 Methodology

DDIMs generalize DDPMs by reparameterizing the forward diffusion process to be non-Markovian, ensuring that the marginal distributions remain unchanged. The generative process is derived by inverting the forward dynamics, enabling deterministic or stochastic sampling depending on the choice of variance parameter σ_t . The approach is connected to solving an underlying ordinary differential equation (ODE), explaining why DDIMs require fewer steps for high-quality sample generation. The acceleration is achieved by using a subset of diffusion timesteps, indexed by $\tau = \{\tau_1, \tau_2, \dots, \tau_S\}$, where $S \ll T$.

3 Key Equations

3.1 Forward Process

The forward diffusion process gradually adds noise to the data:

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (1)$$

The Markovian transition in DDPMs is:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(\sqrt{\frac{\alpha_t}{\alpha_{t-1}}}x_{t-1}, \left(1 - \frac{\alpha_t}{\alpha_{t-1}}\right)I\right). \quad (2)$$

3.2 Loss Function

The denoising loss used for training is:

$$L_\gamma(\varepsilon_\theta) = \sum_{t=1}^T \gamma_t \mathbb{E}_{x_0, \varepsilon} \left[\left\| \varepsilon_\theta^{(t)} (\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon) - \varepsilon \right\|^2 \right], \quad (3)$$

where $\varepsilon_\theta^{(t)}$ is the neural network’s prediction of the noise.

3.3 Reverse Process (Sampling)

To predict the clean sample from a noisy input x_t :

$$f_\theta^{(t)}(x_t) = \frac{x_t - \sqrt{1 - \alpha_t} \varepsilon_\theta^{(t)}(x_t)}{\sqrt{\alpha_t}}. \quad (4)$$

The DDIM sampling step is given by:

$$x_{t-1} = \sqrt{\alpha_{t-1}} \left(x_t - \frac{\sqrt{1 - \alpha_t}}{\sqrt{\alpha_t}} \varepsilon_\theta^{(t)}(x_t) \right) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \varepsilon_\theta^{(t)}(x_t) + \sigma_t \varepsilon_t, \quad (5)$$

where σ_t controls stochasticity, and setting $\sigma_t = 0$ results in a deterministic implicit generative model.

3.4 Accelerated Reverse Process with τ

To make sampling more efficient, instead of running the full T steps, a reduced subset $\tau = \{\tau_1, \tau_2, \dots, \tau_S\}$ is chosen, with $S \ll T$. The updated reverse process follows:

$$x_{\tau_{i-1}} = \sqrt{\alpha_{\tau_{i-1}}} \left(x_{\tau_i} - \frac{\sqrt{1 - \alpha_{\tau_i}}}{\sqrt{\alpha_{\tau_i}}} \varepsilon_\theta^{(\tau_i)}(x_{\tau_i}) \right) + \sqrt{1 - \alpha_{\tau_{i-1}} - \sigma_{\tau_i}^2} \varepsilon_\theta^{(\tau_i)}(x_{\tau_i}) + \sigma_{\tau_i} \varepsilon_{\tau_i}. \quad (6)$$

By selecting fewer steps, this method achieves significant speedups in sample generation while preserving high-level features.

4 Discussion

DDIMs retain the same training objective as DDPMs but allow for accelerated sampling and deterministic inference. By interpreting the generative process as an ODE, DDIMs establish a connection between diffusion models and continuous normalizing flows, providing new insights into efficient generative modeling. The introduction of τ allows for faster sampling by reducing the number of steps needed in the reverse process, striking a balance between speed and image fidelity.